High quality lithium niobate Euler racetrack resonators

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Abstract: We demonstrated Euler racetrack resonators on X-cut thin film lithium niobate with intrinsic quality factors up to 5.53 million and free spectrum range standard deviation 33 times lower than the circular racetrack microcavity.

Thin-film lithium niobate (TFLN) shows great potential to be a platform for integrated photonics due to its outstanding optical properties[1]. Racetrack resonator has been widely used in lithium niobate photonics, particularly in electro-optic comb[2, 3] and modulator[4]. Considering the $\chi^{(2)}$ nonlinearities of lithium niobate, most of these devices prefer using X-cut LNOI to facilitate electrode design[5]. Traditional X-cut racetrack ring encounters two drawbacks. Firstly, the curvature of the straight and the curved part of the racetrack change suddenly, resulting in energy loss and thus decreasing the quality (Q) factor of the cavity. Secondly, due to the anisotropy of the material, the effective refractive index of the light propagating in different directions in the chip varies, resulting in multi-mode crossing, which limits the device's performance[3]. Euler curve with low bending loss and fewer mode crossing is a method that could solve these two problems simultaneously.

In this work, we demonstrated high-quality TFLN racetrack microcavity based on the Euler curve both in simulation and experiment. Our method requires only a single etching step but could achieve a relatively high-quality factor in comparison with the normal circular racetrack cavity with a similar footprint. On the other hand, the mode crossing suppression ability of Euler rings is clearly better than that of circular rings. The free spectrum range (FSR) standard deviation of the Euler cavity is 33 times lower than the circular ring for small footprint cavities (1328 × 60 μm^2) and 5 times lower for large footprint cavities (4355 × 200 μm^2). Not only in the racetrack cavity but the Euler curve could also be used in other TFLN on-chip basic devices such as the mode convertors or delay lines.

Figure 1(a) shows the schematic diagram of a 180° Euler curve which has a decrease (increase) radius of curvature in the upper (lower) part of the curve. We used FDTD 3D simulation to plot the electric field diagram indicating the transmission and scattering of the transverse electric (TE) fundamental mode in Euler and circular curves. Figure 1(b) shows the results for a Euler curve and a circular curve with d of 60 μm. Electric field diagram shows that in the connection part of straight and curve waveguide, more energy leakage occurs at circle curves. Energy leakage in the curve part of racetrack resonator is one of the main reasons of Q factor deterioration.

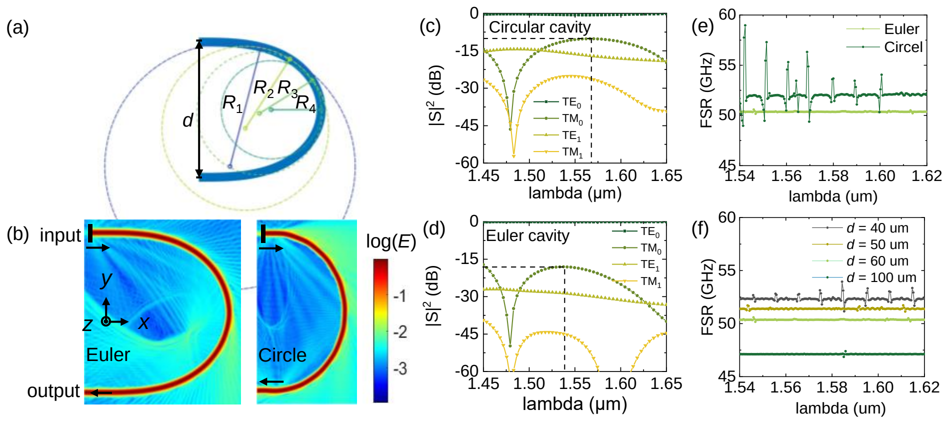

Fig .1 (a) Schematic graph showing the curved part of a Euler cavity with a straight section distance d and changed curvature R. (b) Logarithmic plot of the electric field indicating the light propagation in Euler and circular cavity with equal parameter d. (c) (d)Transmission for TE$_0$, TM$_0$, TE$_1$, TM$_1$ modes when TE$_0$ mode as input for circular (c) and Euler (d) curve. The cross point of dashed lines is the maximum transmission of TM$_0$ mode. (e) FSR changing with wavelength for these two cavities shows the mode crossing suppression of the Euler cavity. (f) FSR changing with wavelength for different parameter d of Euler cavity.

We also calculate the S-parameters at the output port, that is, the transmission of different modes, when TE fundamental mode is input. Comparing Figs. 1 (c) and 1(d), we can see that for TM fundamental mode, the maximum transmission was reduced from -10.2 dB to -18.1 dB by replacing a circular curve with a Euler curve, which proves the excellent performance in mode overlap suppression. A larger transmittance difference between the TE fundamental mode and other modes means that we can get a cleaner TE output mode with less mode crossing.

Experiment results of FSR with respect to wavelength for these two cavities is shown in Fig. 1(e), which was calculated according to the MZI-calibrated wide spectrum. The Euler ring has a smoother FSR curve compared with the circle ring. The FSR standard deviation of the circular (Euler) cavity is 1090 MHz (33 MHz). We change the parameter *d* from 40 um to 100 μm to analyze FSR fluctuation with the change of the device footprint in Fig. 1(f). The FSR standard deviation was 249 MHz, 66 MHz, 33 MHz, and 28 MHz, respectively. A larger footprint tends to have less FSR fluctuation, which is consistent with our simulation results.

We test the large footprint Euler ring that corresponds to a normal circle racetrack ring. Figure 2 (a) and (b) shows Both the loaded (Q_l) and intrinsic quality factor (Q_i) improved by more than 0.5 million than the circle cavity, and the highest Q_i of the Euler cavity of 5.53 million was shown in Fig. 2(c), which demonstrated the high quality of the fabricated Euler cavity. FSR changing with wavelength for Euler and circular cavities is shown in Fig. 2(d). Euler ring has a smoother FSR curve, and the FSR fluctuation standard deviation was reduced from 50.7 MHz to 9.7 MHz. An electro-optic comb with 25GHz modulation frequency was demonstrated in Fig. 2(e), which shows the flat spectrum of the Euler racetrack cavity.

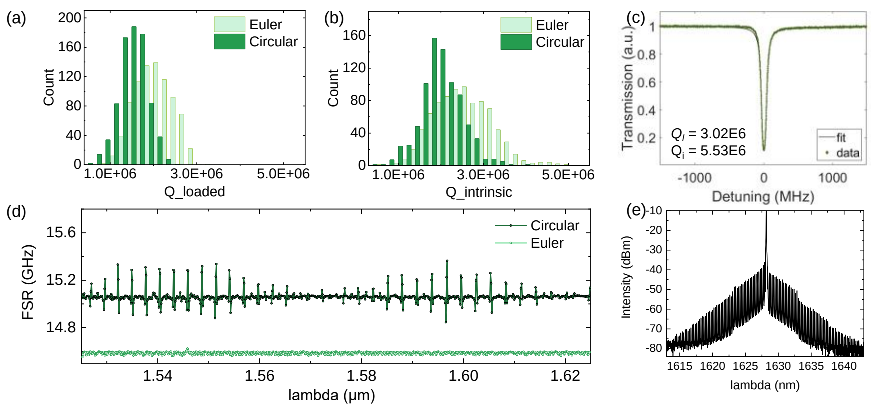

Fig. 2 (a) (b)Histogram graph of Q_l and Q_i of Euler and circular cavity. (c) Single peak fitting plot for Euler cavity with Q_i up to 5.5 million (d) FSR changing with wavelength for Euler and circular cavities. (e) Electro-optic comb with 25 GHz modulation frequency.

In conclusion, we have demonstrated high-quality Euler racetrack microcavity based on TFLN with low FSR fluctuation in comparison with the normal circular racetrack microcavity. Both small-footprint (1328 × 60 um^2) and large-footprint (4355 × 200 um^2) cavities were fabricated. The FSR fluctuation standard deviation was reduced from 1090 MHz to 33 MHz for small-footprint cavities and reduced from 50.7 MHz to 9.7 MHz for large-footprint cavities by using Euler curves. The highest Q_i of the Euler cavity reached 5.53 million. Not only in the racetrack microcavity of electro-optic comb, the Euler curve could be used in the delay-lines or multi-mode waveguide and other situations where the mode needs to be maintained when propagation based on X-cut TFLN.

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